Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

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Project Title

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1. Problem Statement

Ground Glass Opacities (GGOs) appear as faint cloudy areas seen in lung CT scans [1]. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and consumes a large amount of time because GGOs are low contrast areas and spread out gradually in the lungs. This could result in different radiologists interpreting the same CT scan differently, which may affect diagnosis accuracy.

Traditional methods do not have clear and standard ways to measure GGOs. Because of this, it is very hard to accurately track how a disease changes over time or to compare scans taken before and after treatment. Medical images stored in DICOM format are often handled inefficiently, windowing settings (brightness and contrast) vary between hospitals, and many steps are still done manually instead of automatically. These issues make it difficult to interpret CT scans in a consistent and standardized way.

2. Objectives

The main objectives of this project are:

- Improve the quality of the original chest CT scans by performing various image processing steps.
- Accurately segment lung regions using HU based thresholding, morphological operations, and edge filling techniques while removing background artifacts.
- Detect Ground Glass Opacity regions within the segmented lung area using HU based thresholding criteria.
- Test the efficiency of the proposed image processing workflow on a publicly available chest CT dataset.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design and show a simple prototype application which focuses on clear image visualization.

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3. Literature Review

Computed tomography is a cornerstone of pulmonary diagnostics. It reveals subtle parenchymal changes, especially ground glass opacities. Ground glass opacities are faintly hazy areas indicating inflammation or early neoplastic processes. Manual identification of ground glass opacities is the reference standard. Manual identification is time intensive and has inter observer variability. This review surveys major approaches: preprocessing, segmentation, detection, and vessel suppression. It highlights limitations that suggest a need for interactive visualization tools.

3.1. Study 1: Hounsfield Unit Based Lung Segmentation

Early automated lung segmentation used intensity thresholding. Air filled lung tissue has different radiodensity than surrounding structures. Hounsfield unit ranges provide reliable boundaries for tissue types [2]. Simple binary separation distinguishes lung parenchyma from other thoracic components. A 3D approach combined thresholding, dynamic programming, and morphological operations [3]. Dynamic programming helps in regions with faint intensity gradients. Morphological steps remove artifacts like disconnected fragments and incomplete contours. These techniques set early performance standards for segmentation. Accessible implementation recipes followed. One workflow used Hounsfield thresholds of -1000 to -500 HU and morphological closing [4]. Processing lungs independently preserves anatomy in the hilar region. Such pipelines use Python libraries for experimentation and adaptation.

3.2. Study 2: Advanced Segmentation for Diseased Lungs

Disease alters tissue attenuation and texture, challenging thresholding. Interstitial lung disease segmentation combined thresholding and texture maps [5]. Texture maps use gray level co occurrence matrices for spatial arrangement. This method reached 82% overlap with expert outlines. Hybrid strategies integrate fuzzy clustering and morphological refinement. Fuzzy c-means clustering is followed by hole filling, dilation, and reconstruction [6]. Attention is given to juxtapleural nodules blending into the chest wall. Fuzzy clustering handles gradual density transitions better than crisp thresholds. Morphological reconstruction closes gaps without distorting boundaries. These steps help when consolidation or fibrosis obscures lung margins.

3.3. Study 3: Ground Glass Opacity Detection Algorithms

After segmentation, the task is to separate ground glass density from normal tissue. Markov random field models use spatial context among voxels [7]. They achieved 86.94% sensitivity and 94.33% specificity. Morphological vessel removal reduces false positives. Other groups use statistical characterization of candidate regions. First order statistics describe intensity distributions for classification [8]. Entropy is especially discriminative for ground glass texture. Texture oriented classifiers work well for nodular ground glass opacities. Boosted k-nearest-neighbor and density estimation create texture likelihood maps [9]. This method detected all test nodules with few false positives. COVID-19 increased demand for computationally lean methods. Attention thresholding improved Dice scores by 8.9% with low computational cost [10]. It focuses computation on diagnostically salient regions. Volumetric geometric analysis enables new characterization. PointNet++ achieved 98% accuracy and 92% IoU for 3D ground glass regions [11]. Minkowski tensors provide shape descriptors for ground glass opacities. Geometric features allow finer subtyping and may correlate with disease mechanisms.

3.4. Study 4: Pulmonary Vessel Segmentation and Suppression

Pulmonary vessels confound ground glass detection due to similar attenuation. Multiscale Hessian based filtering enhances vessels. A 3D multiscale framework segmented the vascular tree with 96.2–97.0% accuracy [12]. Eigenvalue signatures of tubular geometry are used across scales. Dynamic contour evolution applies geometric constraints. Morphological processing and elliptical fitting use vessel aspect ratios [13]. Edge tracking maintains continuity along vessel paths. Tubular morphology helps reject non vascular detections. Comprehensive pipelines combine thresholding, closing, and Hessian filters. Distance transform processing quantifies vessel proximity [14]. These tools support region aware vessel suppression.

3.5. Gap Analysis

Important limitations remain despite progress. Most GGO detection systems are fully automatic with little user intervention. Radiologists cannot fine tune parameters for individual anatomy or disease. This limits clinical integration. Vessel suppression uses globally uniform settings. Vascular density varies from hilar region to lung periphery. Uniform parameters cause under suppression centrally or tissue loss peripherally. Validation studies focus on overall metrics, not failure modes. Clinicians lack guidance on where improvements are needed. Few methods offer volumetric quantification with slice by slice navigation. Direct correlation between 3D extent and clinical severity is limited. Transparency control for overlay visualization is rarely addressed. Efficiency evaluations focus on algorithmic steps, not end to end workflow. Total processing time determines clinical realism. These limitations motivate interactive visualization tools. Tools should provide adjustable parameters and regionally adaptive processing. Detailed failure mode analysis and volumetric navigation are needed. Overlay transparency control and workflow optimization are important. Such capabilities would bridge research automation and clinical utility.

4. Methodology

The project will consist of 4 main steps:

1. Data Collection and loading
2. Data Preprocessing
3. Lung segmentation
4. GGO visualization and Evaluation

The overall methodology is illustrated in figure 1:

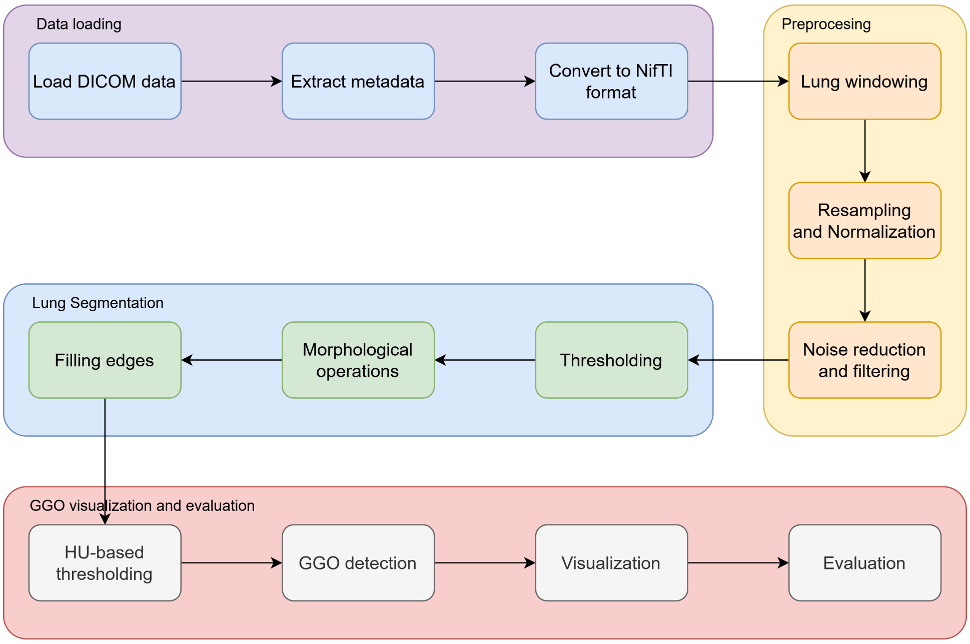

Figure 1: Methodology Overview

4.1. Data Collection and loading

We will use the publically available LIDC-IDRI dataset [15] which contains lung CT scans from the Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI). The dataset contains over 11000 clinical thoracic CT scans in DICOM format. The Python library "pylidc" is the standard tool used to query and visualize this data. The data will be loaded using Python libraries such as "pydicom".

Once the dataset is loaded, we will extract their metadata. The metadata is primarily contained within the DICOM headers of the images and the XML files that hold the radiologist's findings. The LIDC-IDRI dataset provides metadata under three categories:

- **Scan and Patient information:** Includes slice thickness, pixel spacing, intravenous contrast usage

- **Annotations:** Includes type of Lesion and ROI maps
- **Nodule characteristics:** For nodules ¿ 3mm, the metadata includes subjective ratings from 1 to 5 for nine specific features

Once the metadata is extracted, we will convert the DICOM images to standard NifTI format using the "dcm2niix" tool for easier processing in later stages. This conversion helps for compatibility with various medical image processing libraries as DICOM is complex and bulky for analysis.

4.2. Data preprocessing

After loading and extracting metadata, the data will be preprocessed to improve the overall quality of the CT images before segmentation. This step includes:

- Windowing the images to focus on lung tissue
- Resampling and Normalizing pixel intensities
- Removing noise and artifacts

We will apply a lung-specific window level and width to enhance the visibility of lung tissues. This is a medical image convention used to clip the Housenfield range is by choosing a central intensity. Next, resampling will be done to ensure uniform voxel spacing across all scans, which is important for accurate segmentation. This is done because, different scanners may produce images with varying resolutions. Resampling changes the resolution to a standard size such as 1mm x 1mm x 1mm. We will use interpolation methods such as linear or cubic interpolation for this purpose.

Once resampling is done, normalization of pixel intensities will be performed to standardize the intensity values across different scans. This makes mathematical operations more stable and removes dependency on absolute HU values for algorithms. Noise reduction techniques such as Gaussian, Median or Bilateral filtering will be applied to reduce random variations in pixel intensities while preserving important structures like GGO boundaries. Since CT images inherently contain quantum noise, it can create false GGO detections.

4.3. Lung Segmentation

In order to perform Lung segmentation, we will use a combination of intensity based thresholding and morphological image processing techniques. First we will find the real area of the lung in mm^2 by finding the real size of the pixel dimensions. Nex, we will binarize the images using intensity thresholding to separate lung tissues from the background. We expect lungs to be in the Housendfield unit range of [-1000,-300]. To this end, we need to clip the image range to [-1000,-300] and binarize the values to 0 and 1.

Then we need to find the contours of the lung areas. In computer vision, a contour is a set of points that describe a line or area. So for each detected contour we will not get a full binary mask but rather a set with a multiple x and y values. In order to isolate the desired area we will use the marching squares algorithm[]. Using the possible contours, we will extract the lungs using 2 constraints:

- The contour of the lungs must be a closed set
- The contour must have a minimum volume of 2000 pixels to represent the lungs

Using these 2 assumptions, we will convert the contours into a binary mask representing the lung areas by using the "pillow" Python library to draw a polygon connecting the contours After obtaining the binary mask of the lungs, we will save it as a NifTI file and use it for further processing. Finally, we will do element-wise multiplication between the CT image and the lung mask we obtained to get only the lungs.

4.4. GGO Visualization and Evaluation

After segmenting the lungs, we will focus on visualizing the GGO regions within the lung areas. We will assume that if there is a pixel with an intensity value between -500 HU and -700 HU inside the lung area then we will consider it as a GGO area. However, this threshold range may also include vessels within the lung parenchyma. To address this issue, we will apply vessel removal techniques to reduce false positive detections. Since vessels appear as elongated, tubular structures, we will use morphological opening operations using a circular structuring element to remove them. Then, we will set the zeros that resulted from the previous element-wise multiplication to -1000 and finally keep only the intensities that are between -500 and -700 as GGO areas.

Once the GGO area has been identified, we will visualize it by overlaying the detected GGO regions on the original CT images using pseudo-coloring techniques. We will also visualize the original image alongside the segmented GGO area side-by-side using a User Interface. Finally, we will evaluate the effectiveness of the GGO segmentation using manually annotated contours provided in the XML files of the LIDC-IDRI dataset. Back to Table of Contents

5. Dataset / Data Collection

This project utilizes the LIDC-IDRI dataset obtained from The Cancer Imaging Archive (TCIA). The dataset consists of over 1,000 chest CT scans stored in DICOM format. The key statistics of the dataset are summarized as follows:

- **Total cases:** 1,018 patients
- **Total lesions:** 7,371 lesions marked as “nodule” by at least one radiologist
- **Nodules ≥ 3 mm:** 2,669 lesions identified as clinically significant nodules

Images: Clinical thoracic CT scans in DICOM format.
Annotations: Provided in XML files containing the coordinates of nodule boundaries (contours) and associated radiological characteristic ratings.
Tools: The Python library `pylidc` is used to query, process, and visualize the LIDC-IDRI dataset.

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6. Expected Outcomes

The expected outcomes of this project include:

- A Python based image processing pipeline for lung CT analysis
- A GUI application for loading, visualizing, and navigating CT slices
- Accurate lung segmentation masks
- Highlighted Ground Glass Opacity (GGO) regions with color overlays
- Quantitative outputs such as GGO count, area, and estimated volume
- Adjustable HU threshold controls for experimentation
- Side by side visualization of original and processed images

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7. Percentage of Image Processing & Computer Vision Involvement

Core components include DICOM loading, Hounsfield Unit extraction, lung windowing, threshold-based lung segmentation, morphological operations, connected component analysis, flood fill, and GGO detection using HU ranges. Vessel suppression is performed using structural morphology and region-based filtering. Dataset preprocessing and visualization parts of the pipeline are the only components that do not involve image processing or computer vision techniques.

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